MODEL ANSWERS - Generative AI & ML Fundamentals - Mid-Term (Sample)

Answer 1:
Overfitting is when a model learns the training data too closely, including its noise, so it has low training error but high error on new data (poor generalisation). Two remedies: (1) L2 regularisation adds a penalty on large weights, pushing the model towards simpler functions; (2) early stopping halts training when validation loss starts increasing, before the model memorises the training set.

Answer 2:
Each input token embedding x is multiplied by learned matrices W_Q, W_K, W_V to give a query q, key k and value v. The score between token i and token j is q_i.k_j, computed for all pairs as QK^T. Scores are divided by sqrt(d_k) and passed through a softmax over j, giving weights that are positive and sum to 1. The output for token i is the weighted sum of all value vectors: Attention(Q,K,V)=softmax(QK^T/sqrt(d_k))V. Scaling is needed because the dot product of two d_k-dimensional vectors has variance proportional to d_k; large scores make the softmax very peaked (saturated), so gradients become extremely small and training becomes unstable.

Answer 3:
Precision = 40/(40+10) = 0.80. Recall = 40/(40+20) = 0.667. F1 = 2x0.8x0.667/(0.8+0.667) = 1.067/1.467 ~ 0.727. Accuracy = (40+30)/100 = 0.70.

Answer 4:
(a) A GAN trains a generator G and a discriminator D in a minimax game: D learns to tell real from fake, G learns to produce samples that fool D. A diffusion model defines a forward process that adds Gaussian noise over many steps and trains a network to predict the noise (i.e. denoise) at each step. (b) GAN training is notoriously unstable - the two networks must stay balanced and training can diverge or oscillate. Diffusion training is stable because it minimises a simple mean-squared error. (c) GANs often suffer from mode collapse, producing limited variety. Diffusion models cover the distribution better and give more diverse samples. (d) GANs produce a sample in one forward pass, so they are fast; diffusion models need tens to thousands of denoising steps, so sampling is slower (mitigated by DDIM, distillation or consistency models).

Answer 5:
Offline, documents are split into chunks, each chunk is converted to an embedding vector and stored in a vector database. At query time the user question is embedded, the top-k most similar chunks are retrieved, and they are inserted into the prompt so the LLM answers using that context. Failure 1: retrieval misses the relevant passage (poor chunking, vocabulary mismatch) - mitigate with hybrid BM25 + dense search and a cross-encoder re-ranker. Failure 2: the LLM hallucinates or ignores the retrieved context - mitigate by instructing it to answer only from context and cite chunks, abstain when evidence is missing, and run a faithfulness check on the output.